Belief sharing: the standard aggregation corner

`belief_sharing.share_round` lifts the aggregation rule of sec. 8.8 to a colony of categorical sentinel agents.

Belief sharing (part 2)

Each agent has a private sensory outcome and a local posterior over the same shared latent location; it broadcasts the posterior, not its raw observation.

Belief sharing (part 3)

Following the sensory attenuation that the active-inference formulation of belief sharing imposes (Friston et al. 2024) — "agents do not hear themselves" — an agent's heard consensus excludes its own message:

Belief sharing (part 4)

$$q_{-n} \;=\; \mathrm{normalize}(q/t_n)\,, \tag{14}$$

Belief sharing (part 5)

so the round in eq. 14 implements the declared categorical colony-hive-mind mechanism. With the naive fusion rule of eq. 9, eq. 14 is the project's standard log-linear-pool consensus.

Belief sharing (part 6)

Under the explicit shared-support, posterior-log-potential, and fixed-weight assumptions of sec. 8.8, it specializes Eq. 7's message-combination term; it does not reconstruct the complete source protocol.

Belief sharing (part 7)

With the server-side robust rule it yields a hive-mind that can down-weight a contaminated sentinel — an effect the contamination sweep of sec. 10.5 measures rather than assumes, and one that carries no guarantee beyond the recovery limit.

Belief sharing (part 8)

The per-round diagnostics — the post-sharing belief matrix, the global consensus, and the mean surprise and accuracy against a known ground-truth state — are returned by `share_round` and drive Studies 1 and 4.

Belief sharing (part 9)

fig. 6 makes this concrete: three sentinel cards begin with different private categorical views of the nine-cell world, convert those views into local posteriors, and send only those posteriors to a fusion route. The return path is a cavity message,

Belief sharing (part 10)

so the consensus heard by agent n excludes the local posterior q_n.

Belief sharing (part 11)

The figure remains a protocol map rather than a new result; the empirical belief matrix and free-energy comparison remain the evidence surfaces in fig. 13 and fig. 12.

Belief sharing (part 12)

Because eq. 14 calls the aggregation rule of eq. 9 or its robust generalization, the recovery identity of eq. 10 propagates upward: a colony running `share_round` at zero server robustness is bit-identical to a colony running the project's standard log-linear-pool round.

Belief sharing (part 13)

Under the qualified categorical bridge, the pool realizes only the source message-combination specialization; the robust round is a project extension, not a reconstruction of the active-inference ensemble literature (Friston et al. 2024; Heins et al. 2024).

Belief sharing (part 14)

sec. 10.2 reports that communicating colonies reach a mean variational free energy of 13.2190 nats against 16.5298 nats for incommunicado colonies across 480 seeds,

Belief sharing (part 15)

with the per-agent belief matrix before and after a round shown in fig. 13 and the colony comparison in fig. 12.

Belief sharing (part 16)

The honesty boundary of sec. 6.3 carries through the lift unchanged. The robustness that eq. 14 inherits when the colony fuses with the server-side heuristic is the divergence-reweighting device of sec. 8.8,

Belief sharing (part 17)

whose positive property is the naive-recovery limit and whose declared separable objective class has a scoped no-go result.

Belief sharing (part 18)

The per-agent FedGVI bounded-influence result enters the colony only through the rcce/AR client losses of sec. 8.7, under the source theorem's matching assumptions, applied inside each agent's local generalized-Bayes update.

Belief sharing (part 19)

The robustness sweep in sec. 10.5 and the variational supplement in sec. 15 keep the three axes distinct. The federation transport (sec. 16.4) realizes this sharing over queue-backed worker channels;

Belief sharing (part 20)

by Proposition 12, the federation bit-identity result, the consensus is bit-identical to the in-process call, so the channel adds no precision loss while leaving multi-machine network transport as future work.

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